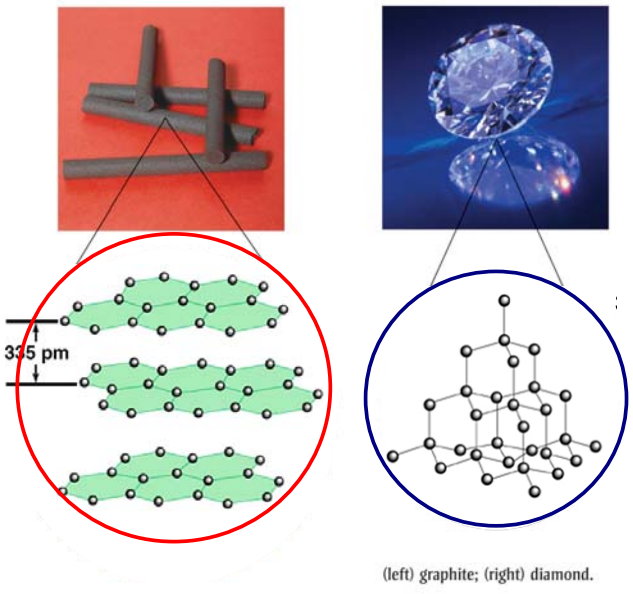
화학특강 (2010년 2학기)
제6장 신물질의 개발과 그 응용

황현석

2010년 10월 28일

Allotropes of Carbon – Graphite and Diamond



(left) graphite; (right) diamond.

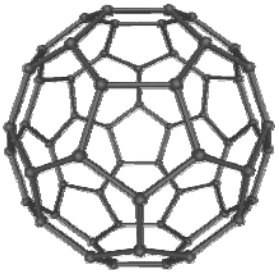
$C_{diamond}(s) + O_2(g) \rightarrow CO_2$ $\Delta G°$=-397 kJ

$C_{graphite}(s) + O_2(g) \rightarrow CO_2$ $\Delta G°$=-394 kJ

$C_{diamond}(s) \rightarrow C_{graphite}(s)$ $\Delta G°$ =? kJ

Kinetic control & Thermodynamic Control

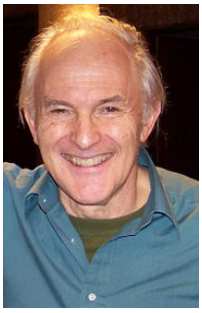
Allotropes of Carbon – Buckminsterfullerene (C_{60})



Buckminsterfullerene (6_{60})
or buckball



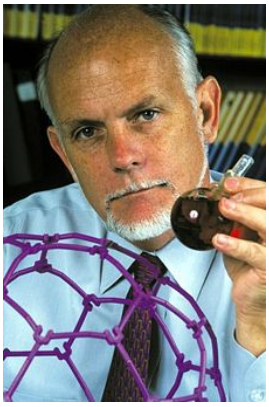
Geodesic dome by
R. Buckminster Fuller



Harold Kroto



Robert Curl

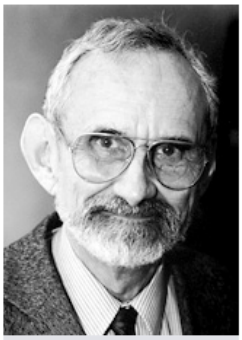


Richard Smalley

KNU

The Nobel Prize in Chemistry 1996

"for their discovery of fullerenes"



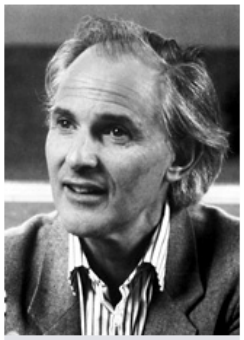
Robert F. Curl Jr.

1/3 of the prize

USA

Rice University
Houston, TX, USA

b. 1933



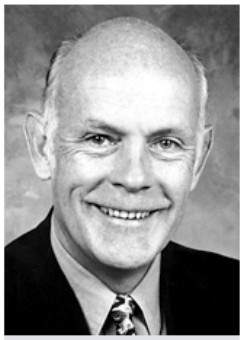
Sir Harold W. Kroto

1/3 of the prize

United Kingdom

University of Sussex
Brighton, United Kingdom

b. 1939



Richard E. Smalley

1/3 of the prize

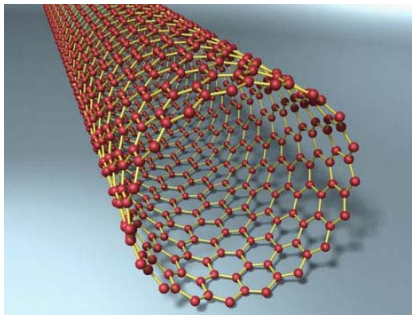
USA

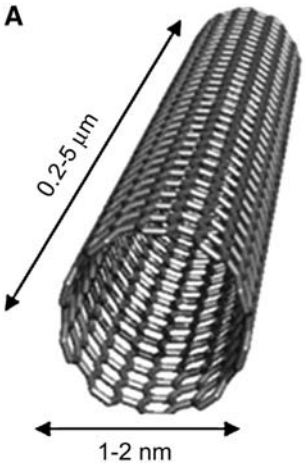
Rice University
Houston, TX, USA

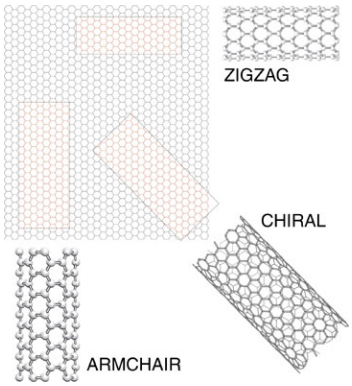
b. 1943
d. 2005

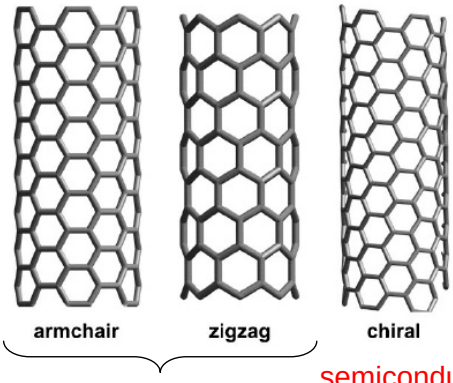
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Allotropes of Carbon – Carbon Nanotube







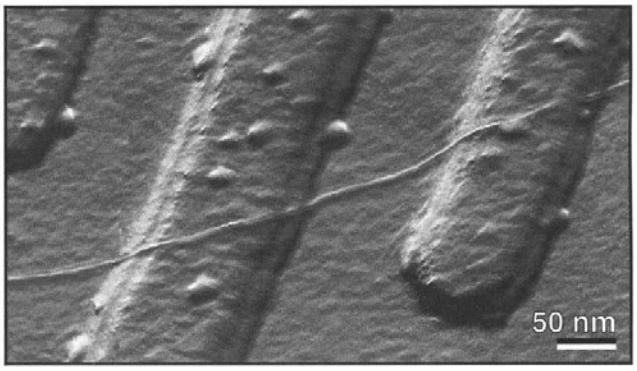


metallic, conductor

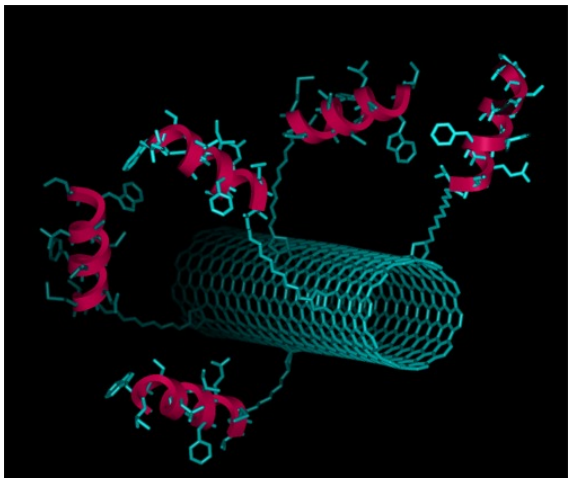
semiconductor

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Applications of carbon nanotubes



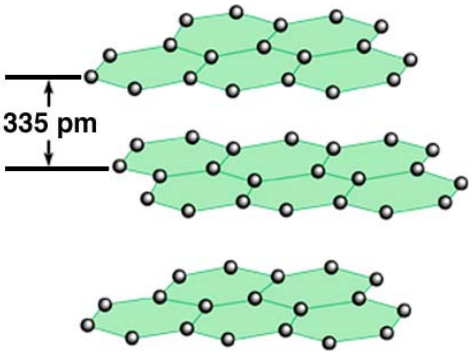
AFM tapping-mode image of a carbon nanotube on top of a Si/SiO_2 substrate (Nature 386, 474 (1997)



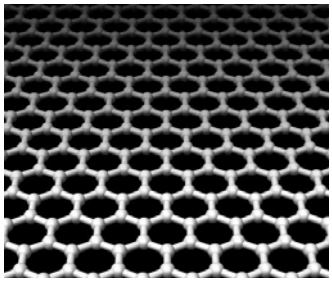
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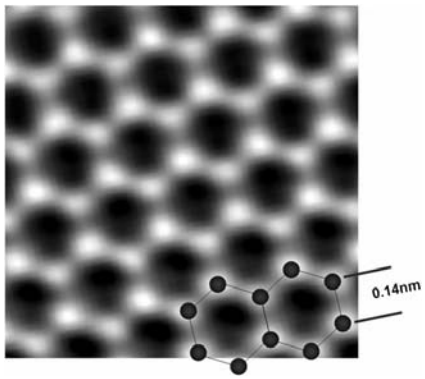
Allotropes of Carbon – Graphene

- Graphite



- Graphene (Science **306**, 666 (2004))





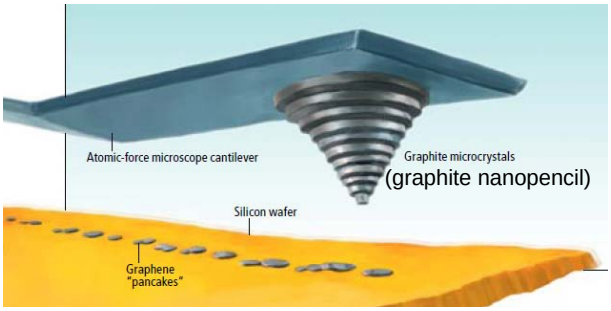
Transmission electron microscope (TEM) image of graphene

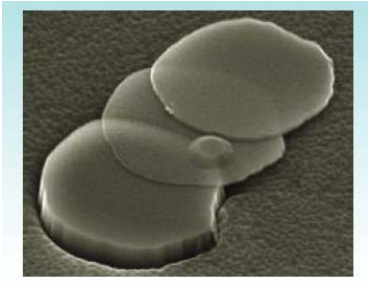
How to Make Graphenes

- Using a plastic sticky tape



- Using a graphite nanopencil

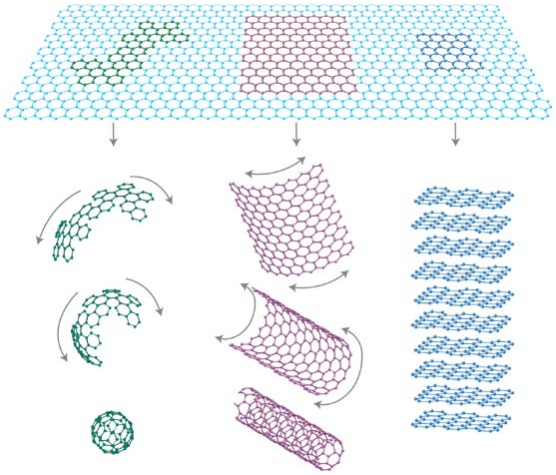


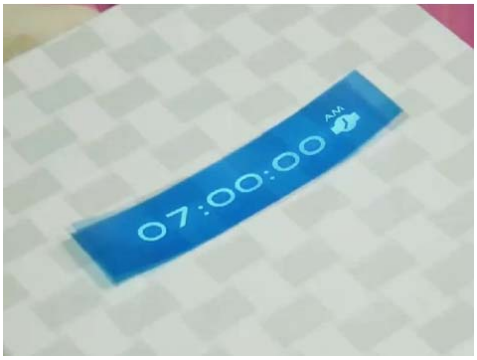


Thin graphene pancakes onto a silicon wafer

Applications of the Graphene

- Mother of most carbon allotropes
- Future applications of graphene

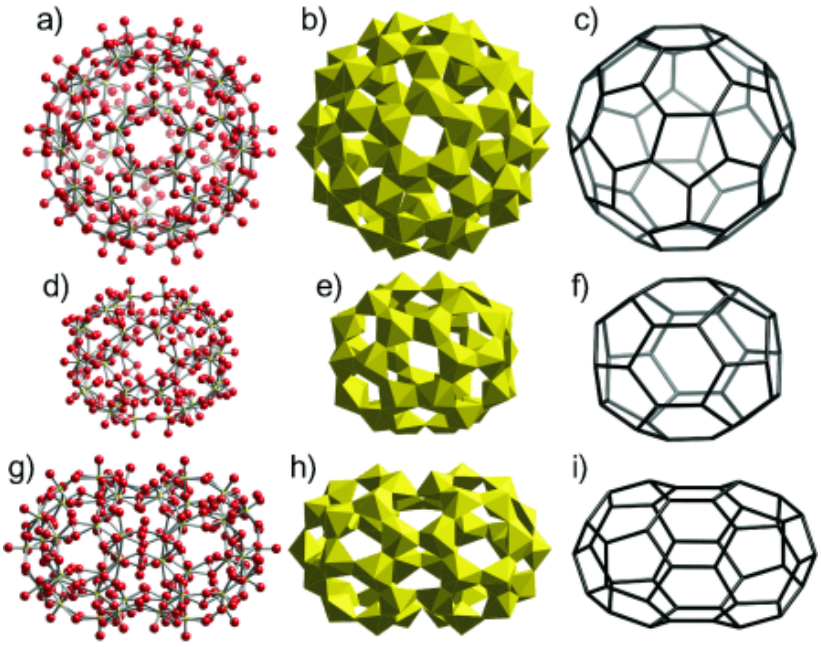




Geim, Nature Mater., **6**, 183 (2007)

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Uranyl Peroxide Fullerene

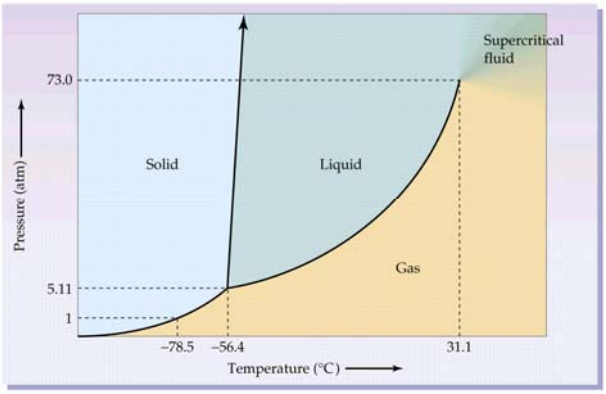


Sigmon et al., Angew. Chem. Int. Ed. **48**, 2737 (2009)

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CO_2 Phase Diagram

- CO_2 Phase Diagram



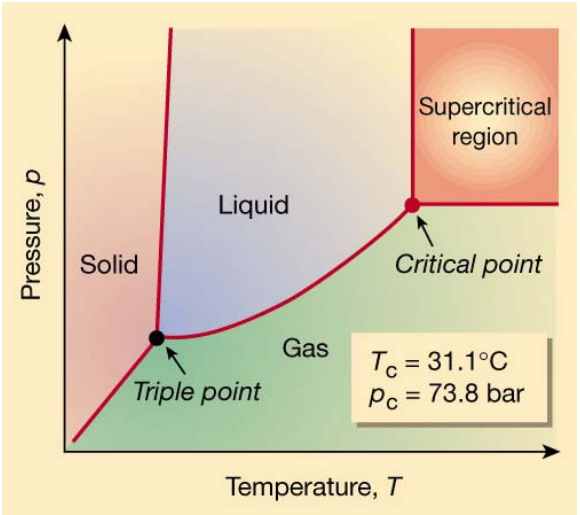


- CO_2 fire extinguisher

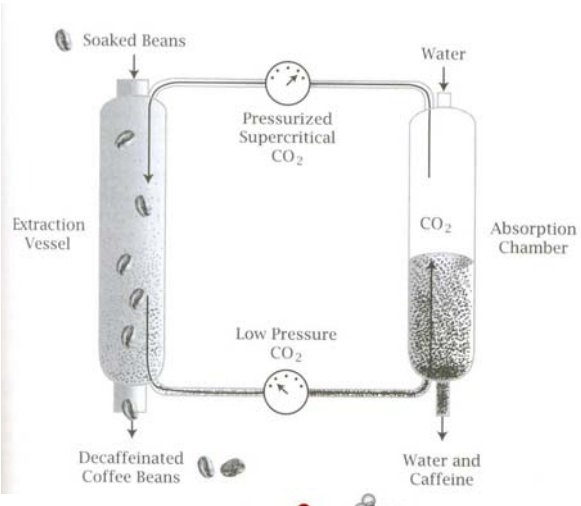


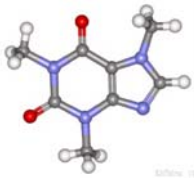
Supercritical CO_2

- Supercritical CO_2 ($scCO_2$)



- Decaffeination using $scCO_2$



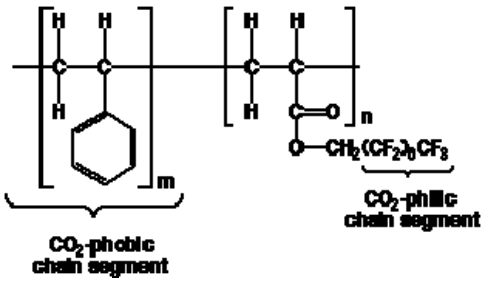


Supercritical CO_2

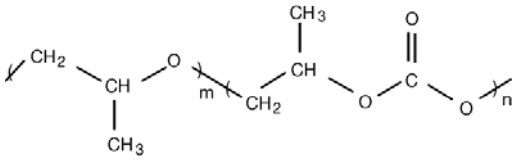
- **Advantages of $scCO_2$**
  - Ecological and economical benefits → no noxious and toxic residues once the solvent has been allowed to evaporate. (green chemistry)
  - A Unique combination of properties from both liquid and gas → low viscosity and high diffusion coefficients of solutes with a density like that of a liquid.
- **Problems in $scCO_2$**
  - Poor solubility of many interesting target substances → Use of solubilizers (or stabilizer)
  - Polysiloxanes and fluorocarbons as solubilizers are expensive.
- **poly(ether_carbonate) copolymers as a solubilizer**
  - Inexpensive.
  - A favorable interaction of the carbonyl group with CO_2 may increase the enthalpy of mixing.

Atkins, Physical Chemistry 8th ed. pp119-120

- **Fluorocarbon copolymer**



- **poly(ether_carbonate) copolymers**

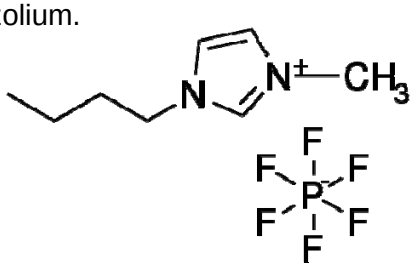


Sarbu et al., Nature 405, 165 (2000)

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Ionic Liquids

- (Room temperature) Ionic liquids: liquids that contains essentially only ions at normal temperature.
- Properties of ionic liquids
  1. Composed of small anions such as BF_4^- (tetrafluoroborate) or PF_6^- (hexafluorophosphate) and large cations such as1-hexyl-3-methylimidazolium or 1-butyl-3-methylimidazolium.



  2. Remain liquids because the bulky, asymmetrical cations do not pack together efficiently with the smaller, symmetrical anions.
  3. Are electrically conductive and have extremely low vapor pressure.

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Ionic Liquids

- Advantages of ionic liquids
  1. Their long liquid ranges, typically from -100°C ~ 200°C.
  2. The environmentally friendly nature.
  3. Can replace organic solvents.
  4. Cations that will attract potentially harmful ions such as mercury, cadmium, uranium, and americium (the latter two found in nuclear waste materials).
  5. Cations that will remove H_2S (which produces SO_2 when gas is burned) and remove CO_2 from the exhaust gases of fossil-fuel-burning power plants.

- Disadvantages of ionic liquids
  1. Their cost – very expensive.

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Ionic Liquids

University of Leicester
DEPARTMENT OF CHEMISTRY
ScioniX

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